

Super-renormalisability and the continuum limit of the scalar Brazovskii functional

TECT verification-first repository · claim A3-RENORMALISATION

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1. Purpose and scope

This note establishes the renormalisation foundation A3: the scalar Brazovskii functional admits a continuum limit with a finite counterterm set, so the “continuum-anchored” language used throughout Sector B refers to a well-defined object. **Scope (narrow, matching A2):** one real scalar field in $d = 3$ with the quartic kernel $K(q) = \mu^2 + Y(q^2 - q_0^2)^2$, $\mu^2 > 0$. The result is PERTURBATIVE super-renormalisability and the continuum limit of the perturbative correlation functions; the non-perturbative (constructive) limit of the full measure is a separate, stronger target (A3_{constructive}), as are the full Class-II action and the $\mu^2 < 0$ branch. Numerical constants are pinned by [codes/foundations/a3_renormalisation_checks.py](#) (6/6 PASS).

2. Setup

The free energy (A1-KERNEL-CONV) is $F_{\text{TECT}}[\phi] = \int [\frac{1}{2}\phi K(-i\nabla)\phi + \frac{\lambda}{4}\phi^4 + \frac{\gamma}{6}\phi^6]$, with bare propagator

$$G(q) = \frac{1}{K(q)} = \frac{1}{\mu^2 + Y(q^2 - q_0^2)^2} \sim \frac{1}{Y q^4} \quad (q \rightarrow \infty), \quad (1)$$

i.e. a q^{-4} ultraviolet decay (assert `propagator_uv_decay_q4`), in contrast to the q^{-2} of a second-order kernel. Vertices are the ϕ^4 (λ) and ϕ^6 (γ) couplings. Regularise by a momentum cutoff Λ (equivalently a lattice spacing $a \sim \Lambda^{-1}$).

3. Power counting

A connected diagram with V vertices, I internal lines and $L = I - V + 1$ loops has, in $d = 3$ with a q^{-4} propagator, superficial degree of divergence

$$D = dL - 4I = 3(I - V + 1) - 4I = 3 - 3V - I. \quad (2)$$

For every connected diagram with at least one vertex and one internal line ($V \geq 1, I \geq 1$),

$$D = 3 - 3V - I \leq 3 - 3 - 1 = -1 < 0 \quad (3)$$

(assert `superficial_degree_all_negative`: the ϕ^4 tadpole, ϕ^4/ϕ^6 self-energies and higher diagrams all have $D \in \{-1, -2, \dots\}$). The dimension $d = 3$ is essential: (2) reads $D = (d - 4)I - dV + d$, which is < 0 for all $V \geq 1, I \geq 1$ only when $d \leq 3$; $d = 4$ is merely marginal.

4. Ultraviolet finiteness (Weinberg)

Every *subdiagram* of any diagram is itself connected with $V' \geq 1, I' \geq 1$, hence has $D' = 3 - 3V' - I' < 0$ by (2). Weinberg’s convergence theorem then gives *absolute* convergence of every loop integral as $\Lambda \rightarrow \infty$ (assert `weinberg_all_subdiagrams_convergent`). Therefore the theory is **ultraviolet-finite**: the perturbative expansion requires NO divergent counterterms. The counterterm set is finite — in fact empty up to the single finite normal-ordering of the tadpole (§5).

5. The tadpole and the continuum limit

The one-loop equal-point contraction is

$$\langle \phi^2 \rangle = \int \frac{d^3 q}{(2\pi)^3} G(q) = \frac{1}{2\pi^2} \int_0^\infty \frac{q^2 dq}{\mu^2 + Y(q^2 - q_0^2)^2}, \quad (4)$$

whose integrand decays as $q^2/(Yq^4) = 1/(Yq^2)$, so the integral converges: $\langle \phi^2 \rangle = 0.767$ at the production parameters, with $|\int^{4\Lambda} - \int^{2\Lambda}| \rightarrow 4.8 \times 10^{-5}$ (assert `tadpole_integral_converges`). Hence the cutoff $\Lambda \rightarrow \infty$ limit is finite. At fixed cutoff, the discretised (lattice) tadpole is an isotropic Riemann sum that converges to the continuum integral as the spacing $a \rightarrow 0$ (assert `discretisation_to_continuum_convergence`: error $3.7 \times 10^{-10} \rightarrow 1.5 \times 10^{-12}$ on refinement, once a resolves the shell width $\sim \sqrt{\mu^2/Y}/q_0$). The two limits combine: the continuum ($a \rightarrow 0$) limit of perturbative correlation functions exists and is cutoff-independent.

6. Infrared finiteness

Under $\mu^2 > 0$, $K(q) \geq \mu^2 > 0$ for all q , so $G(q) \leq 1/\mu^2$ is bounded and no infrared divergence occurs (assert `no_ir_divergence_mu2_positive`). The near-degeneracy of K on the shell $|q| = q_0$ (where $K = \mu^2$) is a finite enhancement, not a divergence.

Theorem (A3, scalar/ $d=3/\mu^2 > 0$). *The scalar Brazovskii functional with the quartic kernel $K(q) = \mu^2 + Y(q^2 - q_0^2)^2$ in $d = 3$, $\mu^2 > 0$, is super-renormalisable: every connected diagram with $V \geq 1, I \geq 1$ has $D = 3 - 3V - I < 0$, so by Weinberg’s theorem the perturbative expansion is ultraviolet-finite and infrared-finite, the counterterm set is finite (empty up to a finite tadpole normal-ordering), and the continuum ($a \rightarrow 0, \Lambda \rightarrow \infty$) limit of the perturbative correlation functions exists and is cutoff-independent.*

7. Devil’s-advocate (three objections)

- (α) “Power counting is only superficial; nested subdivergences could spoil it.” DISMISSED: every subdiagram has $V' \geq 1, I' \geq 1$ hence $D' < 0$ by (2); Weinberg’s theorem requires exactly this (all subdiagrams convergent) and yields absolute convergence. There are no divergent subgraphs to renormalise.
- (β) “This is only perturbative.” VALID with mitigation: the claim is explicitly the perturbative continuum limit (the standard meaning of super-renormalisability). The constructive (non-perturbative measure) limit is flagged as a separate target `A3_constructive`; it is not claimed.
- (γ) “The shell $|q| = q_0$ where $K = \mu^2$ is small could cause trouble.” DISMISSED: $\mu^2 > 0$ keeps $K \geq \mu^2 > 0$, so G is bounded; the shell is a finite enhancement of width $\sim \sqrt{\mu^2/Y}/q_0$, resolved by any a below that scale (§5), with no divergence.

Quantitative sanity check (mandatory): all constants reproduced by [codes/foundations/a3_renormalisation_checks.py](#) (6/6): $D = 3 - 3V - I \in \{-1, -2, -5, -6, -7, -10\}$ for the tabulated diagrams; $K(q)/(Yq^4) \rightarrow 1$; finite tadpole $\langle \phi^2 \rangle = 0.767$; $\min_q K = \mu^2 = 5 \times 10^{-3} > 0$; refinement error $\rightarrow 10^{-12}$.

8. Result footer

Result ID: A3-RENORMALISATION (scalar, d=3, $\mu^2 > 0$)

Precise statement: The d=3 quartic-kernel scalar Brazovskii functional is super-renormalisable: every connected diagram with $V \geq 1, I \geq 1$ has $D = 3 - 3V - I < 0$; by Weinberg the perturbative expansion is UV- and IR-finite, the counterterm set is finite (empty up to a finite tadpole), and the continuum ($a \rightarrow 0$) limit of perturbative correlators exists and is cutoff-independent.

Scope: ONE scalar field, d=3, quartic kernel, $\mu^2 > 0$; PERTURBATIVE. NOT constructive/non-perturbative; NOT full Class-II; NOT $\mu^2 < 0$.

Dependencies: A1-KERNEL-CONV.

Hypotheses: (H1) d=3 with the q^4 kernel [power counting $D < 0$]; (H2) $\mu^2 > 0$ [IR finiteness, $K \geq \mu^2 > 0$].

Evidence grade: PROVED CONDITIONAL (analytic) + EXECUTED (6/6 checks).

Reproduction command: python codes/foundations/a3_renormalisation_checks.py

Expected output: 6/6 PASS; exit 0.

Falsification gate: A connected diagram with $V \geq 1, I \geq 1$ and $D \geq 0$ (would need $d \geq 4$ or a softer-than- q^4 kernel); or a UV/IR divergence at $\mu^2 > 0$; or failure of the $a \rightarrow 0$ Riemann convergence.

Tier before / after: A3 T1 / T6 PROVED CONDITIONAL (perturbative, scalar/d=3/ $\mu^2 > 0$) -- submitted for operator sign-off.

No-overclaim statement: No constructive/non-perturbative continuum limit; no full Class-II; no $\mu^2 < 0$ branch; finiteness is the perturbative statement.

Next required action: Operator sign-off; then A3_constructive (non-pert. measure) and the joint production-Hessian + continuum mainline for BCC stability / mass gap.